

Mark Model v3 — Full Regression Output, Models 1–4

event-scale panel, 5,907 gauge-events, 9 gauges; OLS with gauge-clustered SE

Table 1: **Model 1** (basic wetland effect). Cell: coefficient (clustered SE); stars from clustered p : *** <.01, ** <.05, * <.1.

oprule	log Qp	log peaked.	log TTP	runoff coef	log qdepth	log q99exc
Intercept	4.615*** (0.049)	-9.948*** (0.074)	1.845*** (0.081)	0.089*** (0.003)	-0.493*** (0.082)	0.001*** (0.000)
P (rain)	0.996*** (0.074)	-0.338*** (0.050)	0.250*** (0.022)	0.024*** (0.008)	1.334*** (0.052)	0.001*** (0.000)
W (wetland)	-0.380*** (0.052)	-0.131 (0.183)	0.440*** (0.155)	0.004 (0.005)	-0.249 (0.212)	0.000 (0.000)
log DA	1.223*** (0.066)	-0.005 (0.109)	0.069 (0.117)	0.005 (0.005)	-0.121 (0.140)	-0.000* (0.000)
impervious	1.088*** (0.116)	0.339 (0.429)	-1.053** (0.493)	-0.022 (0.021)	0.748* (0.417)	-0.000 (0.000)
Ksat	0.328*** (0.062)	0.218 (0.134)	-0.350** (0.146)	-0.031*** (0.006)	0.109 (0.164)	-0.000* (0.000)
chan. slope	-0.422*** (0.122)	-0.182 (0.489)	1.210** (0.547)	0.062*** (0.023)	-0.240 (0.476)	0.001*** (0.000)
antecedent	0.256*** (0.046)	0.039* (0.022)	-0.071*** (0.025)	0.020*** (0.003)	0.217*** (0.058)	0.000*** (0.000)
winter	0.348 (0.235)	-0.316*** (0.049)	0.351*** (0.055)	0.103*** (0.012)	0.664*** (0.222)	0.000 (0.000)
N	5907	5907	5907	5907	5907	5907
R^2	0.757	0.329	0.168	0.176	0.680	0.111

Table 2: **Model 2** (+ peak-shaving W×P). Cell: coefficient (clustered SE); stars from clustered p : *** <.01, ** <.05, * <.1.

oprule	log Qp		log peaked.		log TTP		runoff coef		log qdepth		log q99exc	
Intercept	4.651***	(0.047)	-9.932***	(0.075)	1.842***	(0.081)	0.093***	(0.003)	-0.473***	(0.084)	0.001***	(0.000)
P (rain)	0.960***	(0.040)	-0.354***	(0.032)	0.253***	(0.025)	0.020***	(0.003)	1.314***	(0.055)	0.001***	(0.000)
W (wetland)	-0.222***	(0.049)	-0.062	(0.203)	0.424***	(0.164)	0.021***	(0.008)	-0.160	(0.240)	0.000***	(0.000)
W×P	-0.202***	(0.041)	-0.088**	(0.038)	0.020	(0.024)	-0.021***	(0.004)	-0.114*	(0.062)	-0.001***	(0.000)
log DA	1.239***	(0.065)	0.002	(0.108)	0.068	(0.116)	0.007	(0.005)	-0.112	(0.139)	-0.000	(0.000)
impervious	0.922***	(0.134)	0.267	(0.433)	-1.037**	(0.489)	-0.039*	(0.022)	0.655	(0.415)	-0.001**	(0.000)
Ksat	0.251***	(0.059)	0.185	(0.143)	-0.342**	(0.146)	-0.039***	(0.007)	0.066	(0.168)	-0.000***	(0.000)
chan. slope	-0.171	(0.140)	-0.073	(0.501)	1.186**	(0.543)	0.089***	(0.024)	-0.098	(0.484)	0.001***	(0.000)
antecedent	0.262***	(0.044)	0.042*	(0.023)	-0.071***	(0.025)	0.020***	(0.003)	0.220***	(0.057)	0.000***	(0.000)
winter	0.359	(0.231)	-0.312***	(0.047)	0.350***	(0.055)	0.104***	(0.012)	0.670***	(0.221)	0.000	(0.000)
WCB p (W×P)	0.016		0.223		0.480		0.029		0.152		0.027	
N	5907		5907		5907		5907		5907		5907	
R^2	0.768		0.341		0.168		0.189		0.684		0.133	

Table 3: **Model 3** (+ saturation S). Cell: coefficient (clustered SE); stars from clustered p : *** <.01, ** <.05, * <.1.

oprule	log Qp	log peaked.	log TTP	runoff coef	log qdepth	log q99exc
Intercept	4.651*** (0.043)	-9.933*** (0.070)	1.842*** (0.080)	0.093*** (0.003)	-0.473*** (0.071)	0.001*** (0.000)
P (rain)	1.185*** (0.074)	-0.551*** (0.064)	0.162* (0.084)	0.010** (0.004)	1.736*** (0.050)	0.001*** (0.000)
W (wetland)	-0.129*** (0.044)	-0.143 (0.197)	0.387* (0.206)	0.017* (0.009)	0.014 (0.195)	0.000*** (0.000)
W×P	-0.202*** (0.042)	-0.088** (0.038)	0.020 (0.024)	-0.021*** (0.004)	-0.114* (0.062)	-0.001*** (0.000)
S=log(Ve/Sw)	-0.329*** (0.079)	0.286** (0.118)	0.133 (0.142)	0.015*** (0.005)	-0.615*** (0.101)	0.000*** (0.000)
log DA	1.302*** (0.013)	-0.052 (0.100)	0.042 (0.117)	0.004 (0.005)	0.006 (0.094)	-0.000* (0.000)
impervious	0.759*** (0.115)	0.409 (0.450)	-0.971* (0.507)	-0.032 (0.022)	0.350 (0.422)	-0.001** (0.000)
Ksat	0.028 (0.065)	0.378** (0.177)	-0.252 (0.204)	-0.029*** (0.009)	-0.350** (0.172)	-0.000* (0.000)
chan. slope	0.216 (0.147)	-0.410 (0.558)	1.029* (0.625)	0.071*** (0.027)	0.626 (0.521)	0.001*** (0.000)
antecedent	0.264*** (0.044)	0.040* (0.024)	-0.072*** (0.026)	0.020*** (0.003)	0.225*** (0.058)	0.000*** (0.000)
winter	0.357 (0.231)	-0.310*** (0.046)	0.351*** (0.054)	0.104*** (0.012)	0.666*** (0.221)	0.000 (0.000)
WCB p (S=log(Ve/Sw))	0.002	0.365	0.580	0.219	0.217	0.143
N	5907	5907	5907	5907	5907	5907
R^2	0.771	0.351	0.169	0.190	0.691	0.133

Table 4: **Model 4** (+ nonlinear saturation W×S). Cell: coefficient (clustered SE); stars from clustered p : *** <.01, ** <.05, * <.1.

oprule	log Qp	log peaked.	log TTP	runoff coef	log qdepth	log q99exc
Intercept	4.618*** (0.120)	-10.983*** (0.140)	3.164*** (0.092)	0.158*** (0.009)	0.545*** (0.112)	0.001*** (0.000)
P (rain)	1.243*** (0.179)	1.271*** (0.233)	-2.132*** (0.157)	-0.102*** (0.014)	-0.028 (0.169)	-0.000 (0.000)
W (wetland)	-0.201 (0.234)	-2.432*** (0.312)	3.268*** (0.207)	0.158*** (0.021)	2.231*** (0.222)	0.001*** (0.000)
W×P	-0.168* (0.101)	0.974*** (0.142)	-1.317*** (0.091)	-0.087*** (0.011)	-1.142*** (0.088)	-0.001*** (0.000)
S=log(Ve/Sw)	-0.413* (0.247)	-2.380*** (0.341)	3.490*** (0.233)	0.179*** (0.019)	1.967*** (0.290)	0.001*** (0.000)
W×S	-0.049 (0.143)	-1.544*** (0.195)	1.944*** (0.131)	0.095*** (0.012)	1.496*** (0.155)	0.001** (0.000)
log DA	1.297*** (0.023)	-0.226*** (0.032)	0.260*** (0.018)	0.015*** (0.003)	0.174*** (0.021)	0.000 (0.000)
impervious	0.788*** (0.184)	1.333*** (0.221)	-2.134*** (0.133)	-0.089*** (0.016)	-0.545*** (0.128)	-0.001** (0.000)
Ksat	0.040 (0.086)	0.752*** (0.087)	-0.722*** (0.057)	-0.052*** (0.007)	-0.712*** (0.078)	-0.000** (0.000)
chan. slope	0.175 (0.240)	-1.699*** (0.287)	2.652*** (0.177)	0.151*** (0.021)	1.874*** (0.188)	0.002*** (0.001)
antecedent	0.264*** (0.044)	0.039 (0.024)	-0.071*** (0.026)	0.020*** (0.003)	0.226*** (0.058)	0.000*** (0.000)
winter	0.356 (0.230)	-0.315*** (0.045)	0.358*** (0.054)	0.104*** (0.012)	0.672*** (0.221)	0.000 (0.000)
WCB p (W×S)	0.766	0.455	0.311	0.006	0.375	0.162
N	5907	5907	5907	5907	5907	5907
R^2	0.771	0.389	0.210	0.192	0.697	0.133